SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	K. Pujitha	**Reg. No.:**	192524149
Section / Batch:	2nd year (AI&DS)	**Date:**	28/08/2026

Code of Conduct

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Pujitha

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

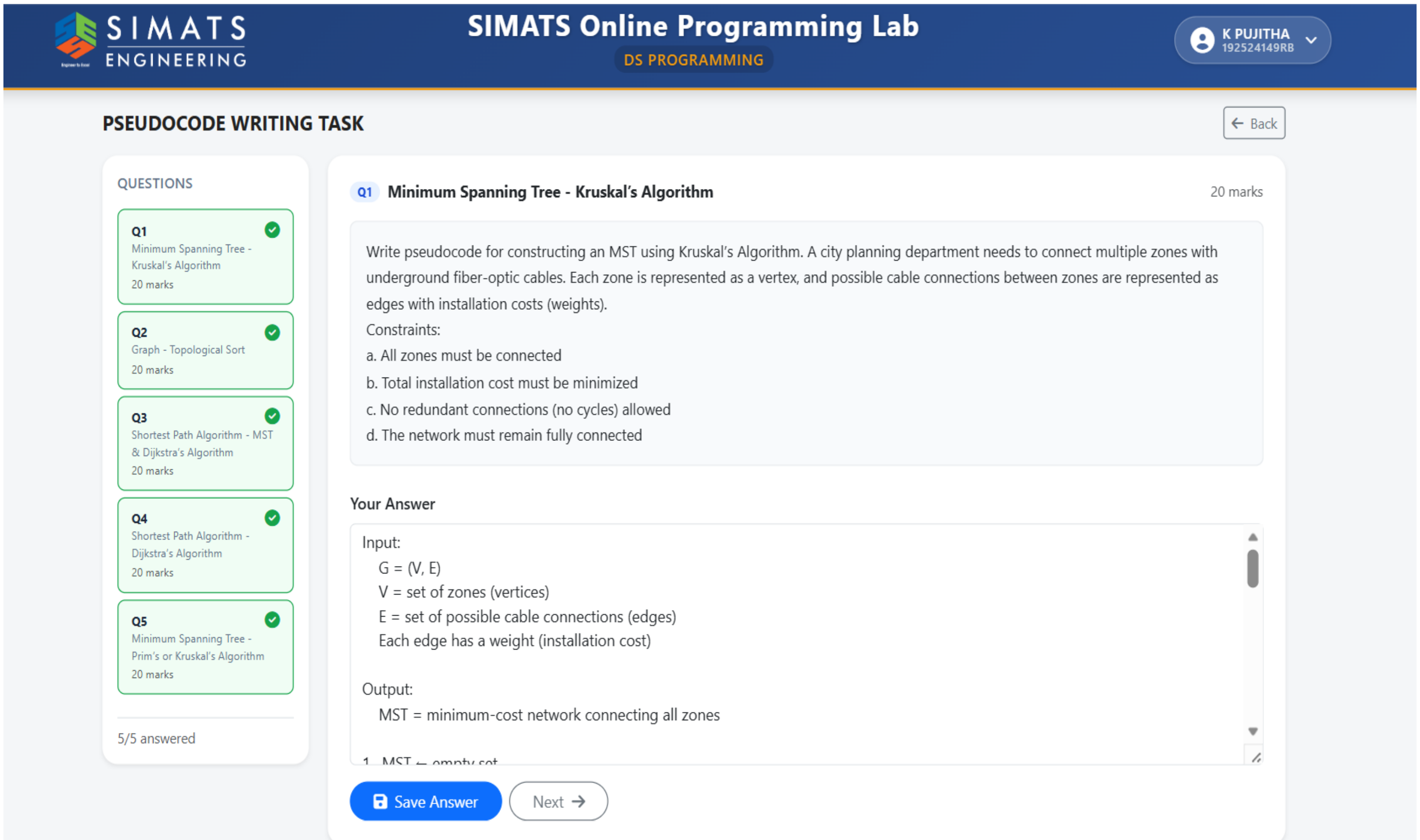
Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

1)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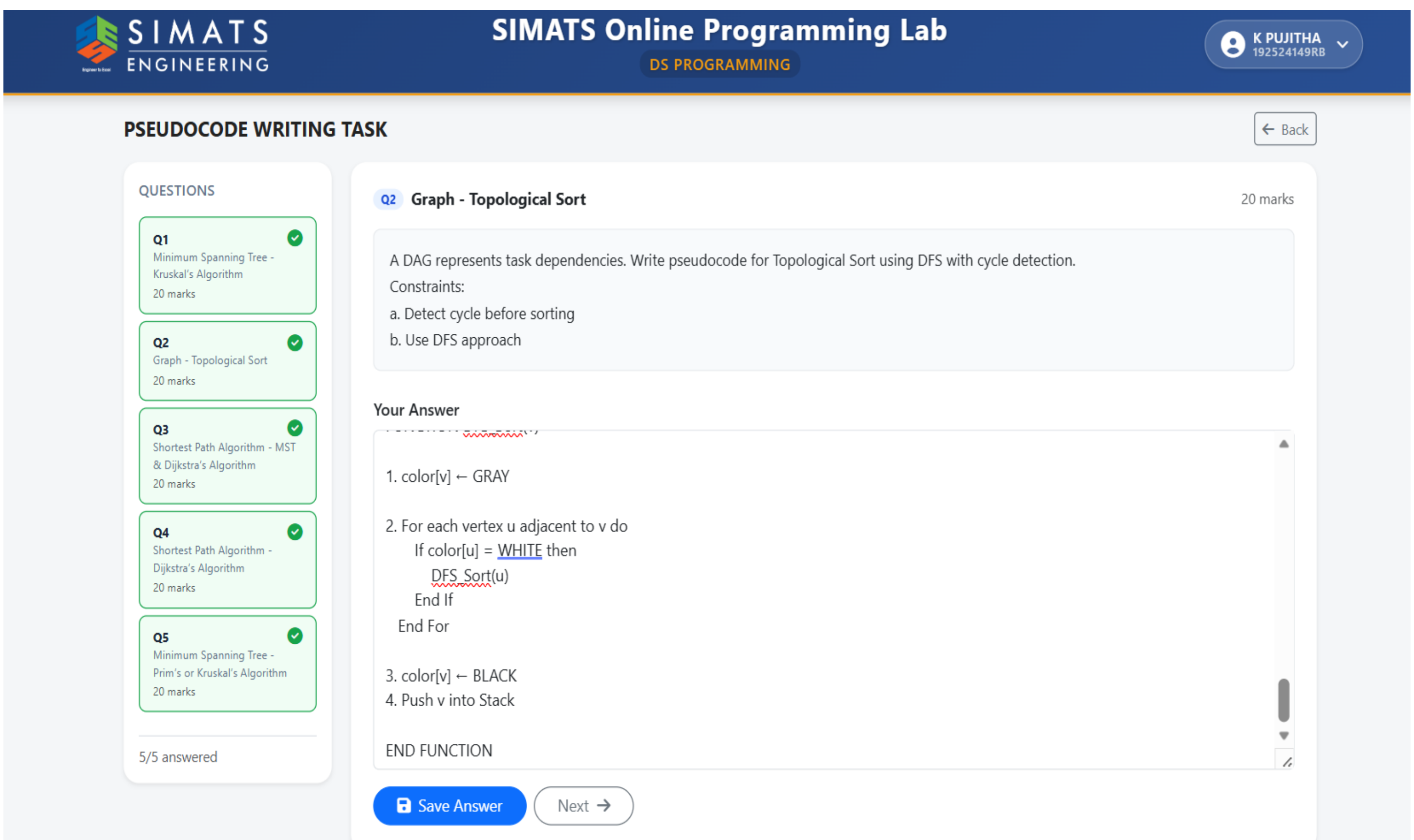
QUESTIONS
Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks
Q2
Graph - Topological Sort
20 marks
Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks
Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q1 Minimum Spanning Tree - Kruskal's Algorithm
20 marks
Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).
Constraints:
a. All zones must be connected
b. Total installation cost must be minimized
c. No redundant connections (no cycles) allowed
d. The network must remain fully connected
Your Answer
Input:
G = (V, E)
V = set of zones (vertices)
E = set of possible cable connections (edges)
Each edge has a weight (installation cost)
Output:
MST = minimum-cost network connecting all zones
Save Answer
Next

2)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20 marks
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Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
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Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q2 Graph - Topological Sort
20 marks
A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach
Your Answer
1. color[v] ← GRAY
2. For each vertex u adjacent to v do
If color[u] = WHITE then
DFS_Sort(u)
End If
End For
3. color[v] ← BLACK
4. Push v into Stack
END FUNCTION
Save Answer
Next

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QUESTIONS

Q1 Minimum Spanning Tree - Kruskal's Algorithm 20 marks

Q2 Graph - Topological Sort 20 marks

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm 20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm 20 marks

5/5 answered

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm 20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

```
        Mark u as visited

        For each neighbor v of u do
            If Distance[u] + Weight(u,v) < Distance[v] then
                Distance[v] ← Distance[u] + Weight(u,v)
                Previous[v] ← u
            End If
        End For
    End While

10. Print shortest path using Previous[]
11. Print Distance[Destination]

END
```

Save Answer Next →

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Q1 Minimum Spanning Tree - Kruskal's Algorithm 20 marks

Q2 Graph - Topological Sort 20 marks

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm 20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm 20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

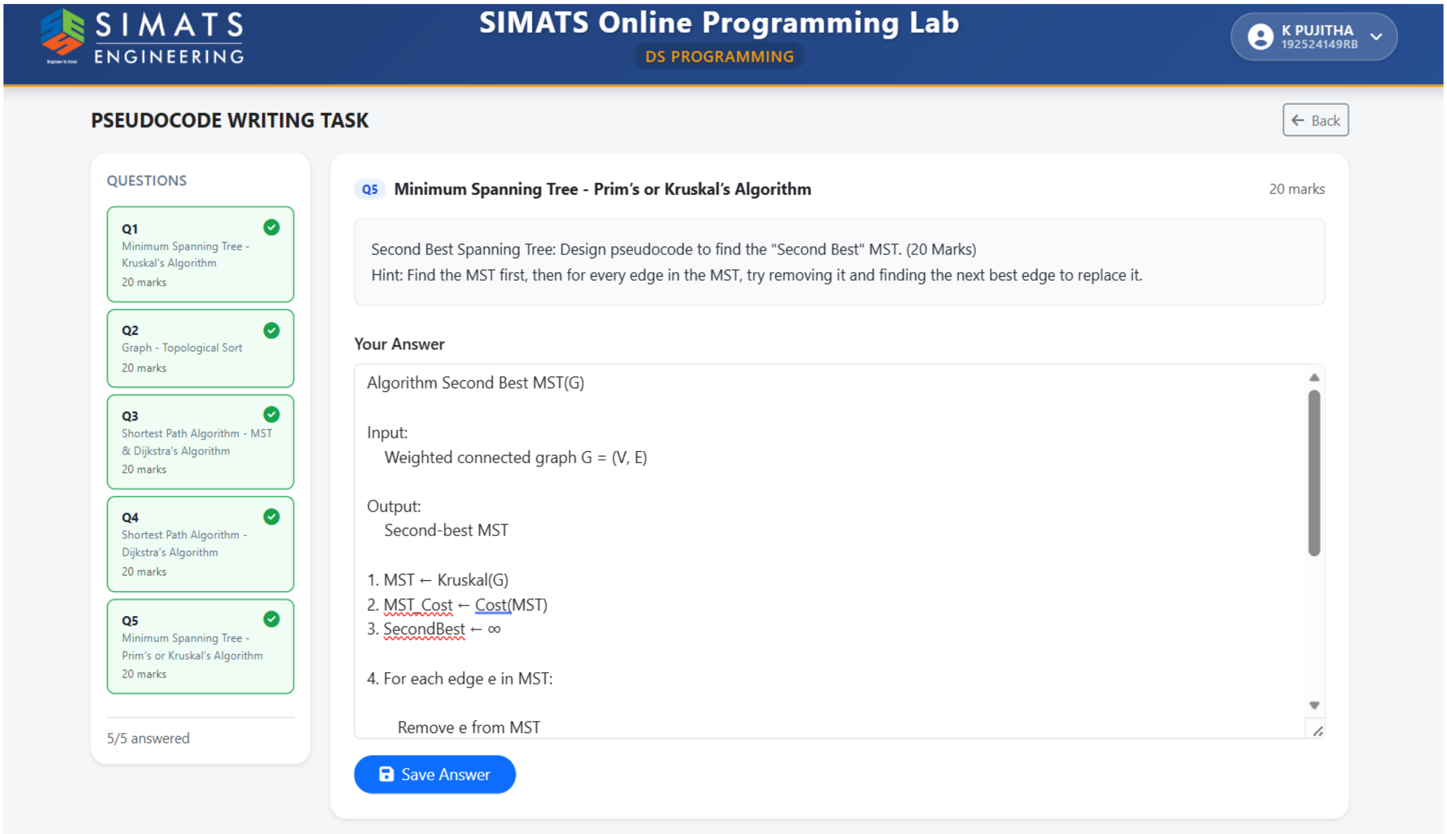
Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
            If newCost < dist[v][used]:
                dist[v][used] = newCost
                Insert (newCost, v, used) into PQ

            // Case 2: Use voucher on this edge
            If used == 0:
                newCost = cost

                If newCost < dist[v][1]:
                    dist[v][1] = newCost
                    Insert (newCost, v, 1) into PQ

5. Answer = min(dist[D][0], dist[D][1])

6. Return Answer
```

Save Answer Next →

5)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Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.
Your Answer
Algorithm Second Best MST(G)
Input:
Weighted connected graph G = (V, E)
Output:
Second-best MST
1. MST ← Kruskal(G)
2. MST_Cost ← Cost(MST)
3. SecondBest ← ∞
4. For each edge e in MST:
Remove e from MST
Save Answer